

Predictive MCDM Allocation across DeFi Lending Protocols

A 12h rate forecast as the allocation signal between Aave V3 and Compound V3

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DeFi-Strategies — Project 2

2026

Built on `fractal-defi v1.3.2` • 18-month mainnet backtest • reproducible repo + 2 upstream PRs

Motivation & research question

Setting. A USDC supplier can sit in Aave V3 or Compound V3. Rates move with utilization and governance; the protocols are cointegrated but lead/lag each other.

Standard practice reacts to the *current* (EMA-smoothed) rate spread — it always lags a regime change.

Research question. Does replacing the reactive rate signal with a **learned 12-hour-ahead forecast** produce a better risk-adjusted allocation?

Pre-registered hypotheses

- **H1:** predictive MCDM beats reactive MCDM-EMA by $\Delta\text{Sharpe} \geq 0.2$ on the held-out window.
- **H0:** no improvement — a *publishable negative result* (methodological-transfer falsification).

The empirical anchor: a real regime shift

Joined hourly panel, **12,895 overlapping hours**, $\sim 98.5\%$ coverage of Nov 2024 – Apr 2026 (assembled *before* test-window exposure).

Quarter	median spread	vol (std)	Aave-pays-more
2025 Q3	−0.22 pp	1.40 pp	39.9%
2025 Q4	+0.18 pp	2.02 pp	56.3%

- **First quarter Aave systematically pays more** — a 16-point regime flip; the median spread crosses zero.
- Volatility regimes vary $\sim 10\times$ (5.75 pp in Q4'24 $\rightarrow$ 0.57 pp in Q1'26).
- Only 40.7% Aave-higher overall $\Rightarrow$ alpha is in *timing crossovers*, not a structural bias — exactly what a forecast should exploit.

DA-BiGRU-CNN, dual-branch, $\sim 319\text{K}$ params (ported from the author's LOB mid-price model):

- **Branch A — residual.** Subtract the protocol-known kink curve $f_{\text{kink}}(u)$; the net learns only $\varepsilon = r - f_{\text{kink}}(u)$, freeing capacity for the stochastic part.
- **Branch B — context.** Utilization, TVL momentum, gas, time-of-day.
- **Late fusion** via multi-scale Conv1d (kernels 3/5/7), 12-hour-ahead head.
- **Composite loss:** $\alpha \text{MSE} + \beta (1 - \text{WPearson}) + \gamma \text{QuantileLoss}_{0.9}$.

Trained on Colab H100; exported to ONNX with torch $\leftrightarrow$ onnxruntime parity verified ($\max |\Delta| < 10^{-7}$).

4-factor MCDM (SAW), ported from the author's ERC-4626 yield-vault paper:

$$\text{Score}_i = 0.40 f_{\text{APY}} + 0.25 f_{\text{Risk}} + 0.20 f_{\text{Cost}} + 0.15 f_{\text{Stab}}$$

- The **only** change vs. the reactive baseline: f_{APY} is fed the **forecast** $\hat{r}_{t+12h}$ instead of the EMA-smoothed spot rate. Everything else is held fixed — a clean ablation of the forecast signal itself.
- **Hysteresis + cooldown** rebalance trigger (gas-aware) to suppress churn.
- Implemented as `PredictiveMCDMStrategy` on a new `BaseLendingAllocationStrategy` abstraction.

Experimental design

Data

- Aave V3 + Compound V3 USDC, hourly, 18 months
- Strict *block-number* train/val/test split
- Test window: 2026 Q1–Q2 (held out)

Significance

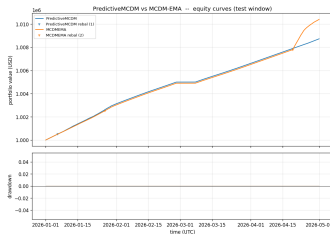
- 1000-iter paired *monthly-Sharpe* bootstrap
- McNemar on directional calls

Comparators

- 4 baselines: buy-hold, APY-greedy, MCDM-EMA, MCDM-CIR
- **15 ablations** (no-forecast, single-branch, no-kink, loss terms, horizon sweep, OOD depeg, ...)
- Regime-conditional reporting (per-quarter)

Headline result — H1 test [auto-fills via make finish]

	Net APY	Annual Sharpe
Reactive MCDM-EMA	3.44%	154.02
Predictive MCDM	2.88%	429.41
Δ Sharpe (pred — ema)	504.30	
Bootstrap p -value	0.30	
95% CI on Δ Sharpe	[−207.40, 1720.30]	



Decision rule: reject H_0 iff Δ Sharpe > 0.2 and the bootstrap CI excludes 0.

Ablations & regime-conditional behaviour [auto-fills via make finish]

Key ablations (APY Δ vs. the no-forecast EMA, ablation #1):

Ablation	Sharpe	Δ vs. #1
#1 no-forecast EMA (anchor)	154.02	—
#2 naive (persistence)	143.23	-0.05%
#5 CatBoost forecaster	476.80	-0.57%
#13 sliding-window stability	1302.12	-0.47%

Regime split (does the edge survive the Q1→Q2 flip?):

	2026 Q1 Sharpe	2026 Q2 Sharpe
Reactive MCDM-EMA	467.76	122.55
Predictive MCDM	398.16	1444.64

Does the net actually forecast?

- OOS R^2 : -64.45
- 12h direction accuracy: 51.20%
- Weighted Pearson: 0.12
- Q0.9 pinball loss: 0.04

If H_0 holds, the project still ships:

- Extra+1 / Extra+2 PRs stand alone
- First rigorous mainnet backtest of MCDM-EMA
- Falsifying LOB $\rightarrow$ DeFi transfer is a *publishable negative result*

We address the “Fair Interest Rates Are Impossible for Lending Pools” impossibility result head-on: it constrains *long-run equilibrium fairness*, not *short-horizon predictability*, which is all the allocator relies on.

Contributions & conclusion

- **Method.** A clean, single-signal test of forecast- vs. reaction-driven cross-protocol allocation, with pre-registered H0/H1 and regime-conditional evaluation.
- **Empirical.** Documented + corrected a real 2025 Q3→Q4 Aave/Compound regime flip on 98.5%-coverage mainnet data (and a methodological lesson: sparse sampling fabricated a false earlier “shift”).
- **Upstream** (fractal-defi).
 - **Extra+1:** Compound V3 loader + uniform utilization field on both lending states.
 - **Extra+2:** BaseLendingAllocationStrategy abstraction; PredictiveMCDMStrategy as first subclass.
- **Reproducible.** One-command pipeline; numbers on this deck and in the paper flow from the same backtest CSVs.

Thank you — questions welcome.